

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER PROJECT FALL/SPRING 202X OR MASTER THESIS FALL/SPRING 202X

BACHELOR OR MASTER IN MATHEMATICS OR MINOR HEADING SUCH AS COURSE TITLE

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1 Introduction

In this project we analyse a model for describing the trajectory of particles of pollutants in groundwater. A successful model for describing underground flows is given by the uncertain Darcy problem. Given a domain D such that its boundary ∂D is divided in three subsets $\Gamma_{\text{in}}, \Gamma_{\text{out}}, \Gamma_N$ such that

$$\Gamma_{\text{in}} \cup \Gamma_{\text{out}} \cup \Gamma_N = \partial D, \quad \Gamma_{\text{in}} \cap \Gamma_{\text{out}} \cap \Gamma_N = \emptyset,$$

the pressure and velocity fields p and u are given by the solution of the following Partial Differential Equation (PDE)

$$\begin{cases} u = -A\nabla p, & \text{in } D, \\ \nabla \cdot u = f, & \text{in } D, \\ p = p_0, & \text{on } \Gamma_{\text{in}}, \\ p = 0, & \text{on } \Gamma_{\text{out}}, \\ \nabla p \cdot n = 0, & \text{on } \Gamma_N, \end{cases} \quad (1)$$

where $\Gamma_{\text{in}}, \Gamma_{\text{out}}$ are the inlet and outlet portions of the boundary of D , and an impermeability condition is imposed on Γ_N .

2 Title of section

Let D be a subset of $\mathbb{R}^d$ and $W(t)$ be a vector of m independent Brownian motions and two functions $f: \mathbb{R}^d \rightarrow \mathbb{R}^d, g: \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$. We consider the following stochastic differential equation (SDE)

$$\begin{cases} dX(t) = f(X(t))dt + g(X(t))dW(t), & 0 < t \leq T, \\ X(0) = X_0, & X_0 \in D. \end{cases} \quad (2)$$

The problem is equipped with two different types of boundary conditions, namely

- i. some item
- ii. another one

2.1 Title of subsection

Look at the appendix. but also let us cite some famous mathematicians, such as [1], [2] and [3].

2.1.1 Title of subsubsection

Some figures of the MA building at EPFL.



(a) Put your sub-caption here



(b) Put your sub-caption here

Figure 1: Put your caption here

3 Background: training dynamics in the Mean ODE limit

We recall in this section the objects and equations that the proof relies on. The forward and backward pass definitions, as well as the gradient-descent update written in terms of the per-sample update map $\mathcal{U}$, are taken verbatim from Chizat [? , Sections 2.1–2.2].

3.1 The ResNet architecture

Consider a ResNet of depth $L \in \mathbb{N}^*$, embedding dimension $D \in \mathbb{N}^*$ and hidden width $M \in \mathbb{N}^*$. For an input $x \in \mathbb{R}^D$, weights $\theta = (z^{j,\ell})_{j,\ell} \in (\mathbb{R}^p)^{M \times L}$ and scaling factor $\alpha > 0$, the forward pass is defined by the recursion

$$\hat{h}_\theta^0(x) = x, \quad \hat{h}_\theta^\ell(x) = \hat{h}_\theta^{\ell-1}(x) + \frac{\alpha}{LM} \sum_{j=1}^M \varphi(\hat{h}_\theta^{\ell-1}(x), z^{j,\ell}), \quad \ell \in \llbracket 1, L \rrbracket, \quad (3)$$

where $\varphi : \mathbb{R}^D \times \mathbb{R}^p \rightarrow \mathbb{R}^D$ is the unit map (e.g. a two-layer-perceptron neuron, a gated linear unit, an attention head, etc.).

3.2 Forward and backward Mean ODE

The infinite-depth limit of (21) is parameterized by an L^2 map $Z : [0, 1] \times \Omega \rightarrow \mathbb{R}^p$ on some abstract probability space $(\Omega, \mathbb{P})$. We view Z as a stochastic process indexed by a depth variable $s \in [0, 1]$; the conditional law of $Z(s)$ represents the distribution of parameters at depth s .

Forward pass. For $x \in \mathbb{R}^D$, the forward Mean ODE is the unique solution $h_Z(\cdot, x) : [0, 1] \rightarrow \mathbb{R}^D$ of

$$h_Z(0, x) = x, \quad \partial_s h_Z(s, x) = \alpha \mathbb{E}[\varphi(h_Z(s, x), Z(s))], \quad s \in [0, 1]. \quad (4)$$

Existence and uniqueness follow from Picard–Lindelöf under Assumption A of [?], which we assume throughout.

Backward pass. For $x, w \in \mathbb{R}^D$, the backward Mean ODE $b_Z(\cdot, x, w) : [0, 1] \rightarrow \mathbb{R}^D$ is the unique solution of

$$b_Z(1, x, w) = w, \quad \partial_s b_Z(s, x, w) = -\alpha \mathbb{E}[D_1 \varphi(h_Z(s, x), Z(s))^\top b_Z(s, x, w)], \quad s \in [0, 1]. \quad (5)$$

Here $D_1 \varphi(x, z)$ denotes the Jacobian of φ with respect to its first argument, and $D_2 \varphi(x, z)$ that with respect to its second argument. The backward pass is the continuous-depth (adjoint method) counterpart of the backpropagation recursion through (21).

3.3 Per-sample update map and GD dynamics

For $z \in \mathbb{R}^p$ and $h, b \in \mathbb{R}^D$, define the *per-sample update map*

$$\mathcal{U}(z, h, b) := -D_2 \varphi(h, z)^\top b \in \mathbb{R}^p. \quad (6)$$

Given a training set $\{(x_i, y_i)\}_{i=1}^n$ and per-sample loss loss_i with adjoint variable $w_{i,k} := \nabla \text{loss}_i(h_{Z_k}(1, x_i))$, the GD dynamics of the Mean ODE in the L^2 geometry reads, with $\xi_0 \sim \mu_0$ and a learning rate $\eta > 0$,

$$Z_0(s) = \xi_0, \quad Z_{k+1}(s) = Z_k(s) + \frac{\eta}{\alpha n} \sum_{i=1}^n \mathcal{U}(Z_k(s), h_{k,i}(s), b_{k,i}(s)), \quad k \in \mathbb{N}, \quad (7)$$

where we use the shorthands $h_{k,i}(s) := h_{Z_k}(s, x_i)$ and $b_{k,i}(s) := b_{Z_k}(s, x_i, w_{i,k})$. The “+” sign is consistent with the definition (24) in which $\mathcal{U}$ already carries a minus sign.

To lighten notation in the proof, we will write the per-sample contribution abstractly as an expectation over a generic data law π :

$$Z_{k+1}(s) = Z_k(s) + \eta \mathbb{E}_{(X,Y) \sim \pi}[\mathcal{U}(Z_k(s), h_k(s, X), b_k(s, X, Y))]. \quad (8)$$

4 Background: equivariance setting

We adopt the framework of [?]. Let G be a compact group with orthogonal representations

$$\rho : G \rightarrow O(\mathbb{R}^D), \quad \hat{\rho} : G \rightarrow O(\mathbb{R}^D), \quad M : G \rightarrow O(\mathbb{R}^p),$$

acting respectively on inputs, outputs, and the parameter space. For $g \in G$, orthogonality gives $\rho_g^\top \rho_g = I_D$ and $M_g^\top M_g = I_p$.

Joint G -equivariance. The unit map $\varphi : \mathbb{R}^D \times \mathbb{R}^p \rightarrow \mathbb{R}^D$ is said to be *jointly G -equivariant* if

$$\rho_g \varphi(x, z) = \varphi(\rho_g x, M_g z), \quad \forall g \in G, \forall (x, z) \in \mathbb{R}^D \times \mathbb{R}^p. \quad (9)$$

This is the standing assumption throughout the present section. It is guaranteed under mild conditions on the activation; see [?, Prop. 11].

Equivariance of derivatives. We will invoke repeatedly the following pointwise identity, which is [?, Prop. 6] specialized to our notation. If φ is Fréchet-differentiable in its first argument and jointly G -equivariant in the sense of (27), then for every $g \in G$,

$$D_1 \varphi(\rho_g x, M_g z) = \rho_g D_1 \varphi(x, z) \rho_g^{-1}. \quad (10)$$

Analogously, if φ is also Fréchet-differentiable in its second argument,

$$D_2 \varphi(\rho_g x, M_g z) = \rho_g D_2 \varphi(x, z) M_g^{-1}. \quad (11)$$

4.1 Standing assumptions on the data and loss

Beyond joint equivariance of φ , the proof requires two structural assumptions on the learning problem.

- **(Equivariance of the data law)** The distribution π on $\mathbb{R}^D \times \mathbb{R}^D$ is G -equivariant:

$$(X, Y) \stackrel{d}{=} (\rho_g X, \hat{\rho}_g Y), \quad \forall g \in G. \quad (12)$$

- **(Invariance of the loss)** The per-sample loss $\ell : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$ is G -invariant:

$$\ell(\rho_g \hat{y}, \hat{\rho}_g y) = \ell(\hat{y}, y), \quad \forall g \in G, \forall \hat{y}, y \in \mathbb{R}^D. \quad (13)$$

Differentiating (31) in $\hat{y}$ and using the orthogonality of ρ_g yields the useful corollary

$$\rho_g \nabla_1 \ell(\hat{y}, y) = \nabla_1 \ell(\rho_g \hat{y}, \hat{\rho}_g y), \quad \forall g \in G. \quad (14)$$

5 Statement and proof

We aim to show that the Mean ODE GD dynamics (26) commutes with the action M_g on the parameter process, provided the data law π is G -invariant and the unit φ is jointly G -equivariant. This is the Neural Mean ODE analogue of [?, Theorem 5].

We denote the pushforward of a law $\mu_s = \text{Law}(Z(s))$ by $\tilde{\mu}_s := M_g \# \mu_s = \text{Law}(M_g Z(s))$, and write $\tilde{Z}(s) := M_g Z(s)$.

5.1 Reduction to equivariance of forward and backward passes

Apply M_g to both sides of the GD update (26): since M_g is linear,

$$M_g Z_{k+1}(s) = M_g Z_k(s) + \eta \mathbb{E}_{(X,Y) \sim \pi} [M_g \mathcal{U}(Z_k(s), h_k(s, X), b_k(s, X, Y))]. \quad (15)$$

We want to show that (33) can be rewritten as

$$\tilde{Z}_{k+1}(s) = \tilde{Z}_k(s) + \eta \mathbb{E}_{(X,Y) \sim \pi} [\mathcal{U}(\tilde{Z}_k(s), \tilde{h}_k(s, X), \tilde{b}_k(s, X, Y))], \quad (16)$$

where $\tilde{h}_k$ and $\tilde{b}_k$ denote the forward and backward passes associated with the transported process $\tilde{Z}_k = M_g Z_k$.

Lemma 5.1 (Equivariance of the update map). *Under (27), for every $g \in G$ and every $(z, h, b) \in \mathbb{R}^p \times \mathbb{R}^D \times \mathbb{R}^D$,*

$$M_g \mathcal{U}(z, h, b) = \mathcal{U}(M_g z, \rho_g h, \rho_g b). \quad (17)$$

Proof. By (24) and orthogonality of ρ_g ,

$$M_g \mathcal{U}(z, h, b) = -M_g D_2 \varphi(h, z)^\top b = -M_g D_2 \varphi(h, z)^\top \rho_g^\top \rho_g b.$$

Taking the transpose of (29) and using $(M_g^{-1})^\top = M_g$ (orthogonality), we get $D_2 \varphi(\rho_g h, M_g z)^\top = M_g D_2 \varphi(h, z)^\top \rho_g^\top$, hence

$$M_g \mathcal{U}(z, h, b) = -D_2 \varphi(\rho_g h, M_g z)^\top \rho_g b = \mathcal{U}(M_g z, \rho_g h, \rho_g b). \quad \square$$

In view of Lemma 8.1, the identity (33) becomes

$$M_g Z_{k+1}(s) = M_g Z_k(s) + \eta \mathbb{E}_{(X,Y) \sim \pi} \left[\mathcal{U} \left(M_g Z_k(s), \rho_g h_k(s, X), \rho_g b_k(s, X, Y) \right) \right]. \quad (18)$$

Comparing (36) with the target (34), the proof reduces to establishing the two identities

$$\rho_g h_Z(s, x) = h_{M_g Z}(s, \rho_g x), \quad \rho_g b_Z(s, x, w) = b_{M_g Z}(s, \rho_g x, \rho_g w), \quad (19)$$

which we prove in the next two subsections.

5.2 Equivariance of the forward Mean ODE

[Equivariance of h_Z] Under Assumption A of [?] and joint G -equivariance (27), for every $g \in G$, every $x \in \mathbb{R}^D$ and every $s \in [0, 1]$,

$$\rho_g h_Z(s, x) = h_{M_g Z}(s, \rho_g x).$$

Proof. Integrating (22) between 0 and s gives

$$h_Z(s, x) = x + \alpha \int_0^s \mathbb{E}[\varphi(h_Z(t, x), Z(t))] dt.$$

Apply ρ_g on both sides and use its linearity, then the joint equivariance (27):

$$\begin{aligned} \rho_g h_Z(s, x) &= \rho_g x + \alpha \int_0^s \mathbb{E}[\rho_g \varphi(h_Z(t, x), Z(t))] dt \\ &= \rho_g x + \alpha \int_0^s \mathbb{E}[\varphi(\rho_g h_Z(t, x), M_g Z(t))] dt. \end{aligned}$$

Set $A(s) := \rho_g h_Z(s, x)$. Then

$$A(s) = \rho_g x + \alpha \int_0^s \mathbb{E}[\varphi(A(t), M_g Z(t))] dt. \quad (20)$$

On the other hand, set $B(s) := h_{M_g Z}(s, \rho_g x)$. By the very definition (22) of the forward Mean ODE (with parameter process $M_g Z$ and initial condition $\rho_g x$),

$$B(s) = \rho_g x + \alpha \int_0^s \mathbb{E}[\varphi(B(t), M_g Z(t))] dt. \quad (21)$$

Thus A and B solve the same integral equation, with the same initial condition. The right-hand side defines a continuous, Lipschitz-in-the-state vector field (Assumption A; see also Lemma 5.1 of [?]); by the Picard–Lindelöf theorem the solution is unique, hence $A \equiv B$, i.e.

$$\rho_g h_Z(s, x) = h_{M_g Z}(s, \rho_g x). \quad \square$$

5.3 Equivariance of the backward Mean ODE

[Equivariance of b_Z] Under the same assumptions as Proposition 8.2, for every $g \in G$, every $x, w \in \mathbb{R}^D$ and every $s \in [0, 1]$,

$$\rho_g b_Z(s, x, w) = b_{M_g Z}(s, \rho_g x, \rho_g w).$$

Proof. Integrating (23) between s and 1 and using the terminal condition $b_Z(1, x, w) = w$,

$$b_Z(s, x, w) = w - \alpha \int_s^1 \mathbb{E} \left[D_1 \varphi(h_Z(t, x), Z(t))^\top b_Z(t, x, w) \right] dt.$$

Apply ρ_g on both sides; inserting $\rho_g^\top \rho_g = I_D$ inside the expectation and using (28) (in transposed form, $\rho_g D_1 \varphi(h, z)^\top \rho_g^\top = D_1 \varphi(\rho_g h, M_g z)^\top$),

$$\begin{aligned} \rho_g b_Z(s, x, w) &= \rho_g w - \alpha \int_s^1 \mathbb{E} \left[\rho_g D_1 \varphi(h_Z(t, x), Z(t))^\top \rho_g^\top \rho_g b_Z(t, x, w) \right] dt \\ &= \rho_g w - \alpha \int_s^1 \mathbb{E} \left[D_1 \varphi(\rho_g h_Z(t, x), M_g Z(t))^\top \rho_g b_Z(t, x, w) \right] dt. \end{aligned}$$

Using Proposition 8.2 to replace $\rho_g h_Z(t, x)$ by $h_{M_g Z}(t, \rho_g x)$,

$$\rho_g b_Z(s, x, w) = \rho_g w - \alpha \int_s^1 \mathbb{E} \left[D_1 \varphi(h_{M_g Z}(t, \rho_g x), M_g Z(t))^\top \rho_g b_Z(t, x, w) \right] dt.$$

Set $A(s) := \rho_g b_Z(s, x, w)$ and $B(s) := b_{M_g Z}(s, \rho_g x, \rho_g w)$. The previous identity reads

$$A(s) = \rho_g w - \alpha \int_s^1 \mathbb{E} \left[D_1 \varphi(h_{M_g Z}(t, \rho_g x), M_g Z(t))^\top A(t) \right] dt, \quad (22)$$

while by definition (23) of the backward Mean ODE applied to the parameter process $M_g Z$ with initial datum $\rho_g x$ and terminal value $\rho_g w$,

$$B(s) = \rho_g w - \alpha \int_s^1 \mathbb{E} \left[D_1 \varphi(h_{M_g Z}(t, \rho_g x), M_g Z(t))^\top B(t) \right] dt. \quad (23)$$

Equations (40) and (41) are the same linear integral equation in the unknown with terminal datum $\rho_g w$. By Picard–Lindelöf (applied after time-reversal $s \mapsto 1 - s$, so as to bring the equation to the standard initial-value form already used in Proposition 8.2), the solution is unique, so $A \equiv B$:

$$\rho_g b_Z(s, x, w) = b_{M_g Z}(s, \rho_g x, \rho_g w). \quad \square$$

5.4 Conclusion: equivariance of the GD update

To bring all ingredients together, let us re-introduce the dependence of the backward pass on the loss. Writing

$$w_Z(X, Y) := \nabla_1 \ell(h_Z(1, X), Y),$$

the GD update (26) reads more explicitly

$$Z_{k+1}(s) = Z_k(s) + \eta \mathbb{E}_{(X, Y) \sim \pi} [\mathcal{U}(Z_k(s), h_{Z_k}(s, X), b_{Z_k}(s, X, w_{Z_k}(X, Y)))]. \quad (24)$$

Plugging the equivariance identities (37) into (36) yields

$$M_g Z_{k+1}(s) = M_g Z_k(s) + \eta \mathbb{E}_{(X, Y) \sim \pi} [\mathcal{U}(M_g Z_k(s), h_{M_g Z_k}(s, \rho_g X), b_{M_g Z_k}(s, \rho_g X, \rho_g w_{Z_k}(X, Y)))]. \quad (25)$$

It remains to absorb the residual ρ_g factors inside the expectation. We do so in two steps: first an equivariance identity for w , then a change of variable in the expectation.

Lemma 5.2 (Equivariance of the gradient of the loss). *Under joint equivariance of φ and loss invariance (31),*

$$\rho_g w_Z(X, Y) = w_{M_g Z}(\rho_g X, \hat{\rho}_g Y), \quad \forall (X, Y) \in \mathbb{R}^D \times \mathbb{R}^D. \quad (26)$$

Proof. By (32) applied at $\hat{y} = h_Z(1, X)$, $y = Y$,

$$\rho_g w_Z(X, Y) = \rho_g \nabla_1 \ell(h_Z(1, X), Y) = \nabla_1 \ell(\rho_g h_Z(1, X), \hat{\rho}_g Y).$$

By Proposition 8.2 applied at $s = 1$, $\rho_g h_Z(1, X) = h_{M_g Z}(1, \rho_g X)$, hence

$$\rho_g w_Z(X, Y) = \nabla_1 \ell(h_{M_g Z}(1, \rho_g X), \hat{\rho}_g Y) = w_{M_g Z}(\rho_g X, \hat{\rho}_g Y). \quad \square$$

Substituting (44) into (43) gives

$$M_g Z_{k+1}(s) = M_g Z_k(s) + \eta \mathbb{E}_{(X, Y) \sim \pi} [\mathcal{U}(M_g Z_k(s), h_{M_g Z_k}(s, \rho_g X), b_{M_g Z_k}(s, \rho_g X, w_{M_g Z_k}(\rho_g X, \hat{\rho}_g Y)))]. \quad (27)$$

Finally, perform the change of variable $(X', Y') := (\rho_g X, \hat{\rho}_g Y)$. By the equivariance of π (30), $(X', Y') \sim \pi$ as well, so the expectation in (45) can be rewritten as

$$\tilde{Z}_{k+1}(s) = \tilde{Z}_k(s) + \eta \mathbb{E}_{(X', Y') \sim \pi} [\mathcal{U}(\tilde{Z}_k(s), h_{\tilde{Z}_k}(s, X'), b_{\tilde{Z}_k}(s, X', w_{\tilde{Z}_k}(X', Y')))], \quad (28)$$

where $\tilde{Z} := M_g Z$. This is precisely the GD update equation (42) applied to the transported parameter process $\tilde{Z}$. We have therefore shown that the action of M_g on the parameter process commutes with the Mean ODE GD dynamics. $\square$

Remark 5.1 (Equivariance of the dynamics versus preservation of SI). *The result above is a structural property of the Mean ODE GD dynamics: it asserts that the diagram*

$$\begin{array}{ccc} Z_k & \mapsto & Z_{k+1} \\ \downarrow M_g & & \downarrow M_g \\ \tilde{Z}_k & \mapsto & \tilde{Z}_{k+1} \end{array}$$

commutes. No assumption is required on the initial distribution μ_0 for this conclusion.

Strong invariance of the initialization, i.e. $\mu_0 \in \mathcal{P}(E^G)$, is a logically separate input that is used afterwards: it ensures $M_g Z_0 = Z_0$ almost surely, equivalently $\tilde{Z}_0 = Z_0$ in distribution. Combined with the commutativity above, an immediate induction yields $\tilde{Z}_k = Z_k$ in distribution for every $k \geq 0$, i.e. $M_g \# \mu_k = \mu_k$ for every $g \in G$. Hence the dynamics preserves the space of SI parameter laws $\mathcal{P}(E^G)$. This is the Neural Mean ODE analogue of [?, Theorem 5].

The same scheme yields the analogue of [?, Theorem 4] on the preservation of weakly-invariant (WI) laws, by replacing “ $\mu_0 \in \mathcal{P}(E^G)$ ” by “ $\mu_0 \in \mathcal{P}^G(\mathcal{Z})$ ”.

6 Conclusion

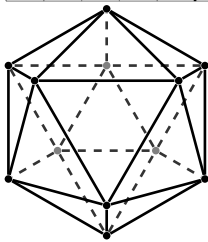
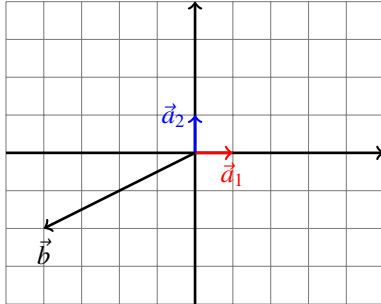
In this project we did a lot of things.

Appendix A Symbols

In the The Comprehensive L^AT_EX Symbol List, one can find a **lot** of symbols. Such as $\therefore \boxminus \phi \leftrightarrow \leftrightsquigarrow$ or even some non-maths ones $\text{\textcircled{A}}$ $\text{\textcircled{B}}$.

Appendix B Using tikz for drawing

Tikz can help you draw some nice things. but other tools exist as well. Take a look at Tikz pour l'impatient
Some examples:



References

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